The study interacts only with commercial AI systems through standard interfaces. No human participants, personal data, or sensitive content were involved. As such, this work does not constitute human-subjects research and is exempt from IRB review.

2.9 Data Availability

Data, prompts, and analysis code are available on the OSF project page at: https://osf.io/dc24t/.

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3. Results

3.1 Baseline difficulty check

The 100-item dataset is challenging but not adversarial. Using the six Baseline models as a proxy for difficulty:

- 11 items were solved by all 6 Baselines.
- 30 items by 5/6 Baselines.
- 46 items by 4/6 Baselines.
- 11 items by 3/6 Baselines.
- 2 items by only 2/6 Baselines.
- 0 items were not solved or solved by only 1 Baseline model.

Thus, most questions are "easy" for current-generation models, but a small tail of harder items remains, especially for weaker prior-generation models. This provides a non-trivial forecasting challenge: Predictors can't just assume everyone is always right.

3.2 RQ1 / H1: Do incentives improve accuracy?

Result: Incentives improved accuracy modestly, but not at conventional significance levels in this small pilot.

Aggregating across all Predictors and repetitions (9 runs per condition, 5,400 predictions per condition):

- **Control accuracy:** 79.1%
- **Incentive accuracy:** 81.5%
- **Difference:** +2.4 pp (Incentive > Control)

We conducted the preregistered statistical analyses on the 9 runs (three per Predictor per condition). An independent-samples t-test comparing run-level mean prediction accuracy showed a modest but statistically non-significant advantage for the Incentive condition (M = 81.5%, SD = 2.2%) over Control (M = 79.1%, SD = 3.2%), t(16) = 1.81, p = .089 (two-tailed), with a large standardized effect size (Cohen's d = 0.86), though the 95% CI [−0.20, 1.90] spans zero, indicating the true effect could plausibly range from slightly negative to very large given this sample size.

Table 2. Statistical test results for preregistered hypotheses.

Analysis	Statistic	p-value	Effect size
H1: Overall accuracy (Incentive vs Control)	t(16) = 1.81	.089	d = 0.86
H2: Learning improvement (R4 − R1)	t(16) = 2.87	.011*	d = 1.35
Round × Condition interaction	z = 3.06	.002**	β = 0.03

*Note. All p-values are two-tailed (α = .05 per preregistration). H1 and H2 use independent-samples t-tests on run-level means (n = 9 per condition); the Round × Condition entry reports the Wald z and slope from the mixed-effects model with Predictor as a random effect; *p < .05. **p < .01.*

Taking the simple average of the three Control runs vs the three incentive runs, the improved forecasting of the Incentive case was 0.8 pp for Mistral, 2.3 pp for Qwen, and 4.0 pp for Grok. Because this is a small-n exploratory pilot, these tests should be treated as effect-size estimates rather than confirmatory hypothesis tests, consistent with the preregistered goals.

Table 3. Prediction accuracy and learning by condition.

Predictor	Condition	Accuracy (%)	R4 − R1 (pp)
Grok	Control	78.3	0.0
Grok	Incentive	82.2	10.6
Mistral	Control	79.6	4.3
Mistral	Incentive	80.3	11.1
Qwen	Control	79.7	4.4
Qwen	Incentive	81.9	14.2
All	**Control**	**79.1**	**2.9**
All	**Incentive**	**81.5**	**12.0**

Note. R4 − R1 = Round 4 accuracy minus Round 1 accuracy. pp = percentage points. n = 3 runs per predictor per condition.

Given the small sample and non-independent error structure (multiple predictions per question and per model), these numbers should be interpreted **descriptively**, not as a definitive statistical win for H1. Informally, the data suggest:

- LLMs are pretty good at knowing what other LLMs know (≈80% accuracy).
- Simple financial framing yields a slight, noisy boost in accuracy on the order of a couple of percentage points in this setup.

3.3 RQ2 / H2: Learning across rounds

Both conditions showed improvement from Round 1 to Round 4, with much larger gains coming in the Incentive case.

- **Control:** Accuracy improved by 2.9 pp by the fourth round.
- **Incentive:** Accuracy climbed more smoothly from roughly **~75%** in Round 1 to **~87%** in Round 4.

Table 4. Accuracy by round and condition.

Round	Control (%)	Incentive (%)	Difference (pp)
1	79.4	75.3	−4.1
2	74.7	79.1	+4.5
3	80.0	84.4	+4.4
4	82.3	87.2	+4.9

Note. Values are means across 9 runs per condition.

A mixed-effects model with Predictor as a random effect revealed a significant Round × Condition interaction ($\beta = 0.028$, $z = 3.06$, $p = .002$). Separate models showed that accuracy increased 4.12 pp per round in the Incentive condition ($p < .001$) versus 1.37 pp in Control ($p = .057$)—a threefold difference in learning rate. A direct comparison of Round 4 - Round 1 improvement confirmed this pattern: Incentive runs improved by 12.0 pp on average versus 2.9 pp for Control, $t(16) = 2.87$, $p = .011$ (two-tailed), $d = 1.35$.

Diving deeper into the logs suggests a plausible story: